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Leveraging STARA competencies and green creativity to boost green organisational innovative evidence: A praxis for sustainable development

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Abstract

Radical technological advancements and the relentless progression of climate change compel organisations to ensure their workforce consistently exercise their creativity toward innovative green initiatives. These endeavours are essential to achieve the United Nations' (UN) sustainable development goals (SDGs). To do so, organisations require competencies fundamental to smart technologies, artificial intelligence, robotics and algorithms (STARA). STARA competencies are relevant for leaders to bolster green organisational innovative evidence (GOIE). GOIE can help to attract potential investors keen on advancing the UN's SDG agenda on environmental sustainability. However, eclipsed by a volatile environment, and despite the green innovation potential of several manufacturing organisations, investors are reluctant to invest and commit funds without evidence of green innovation. We therefore, investigate how leader STARA competence (LSC), green creativity components (task motivation, creativity skills and expertise) and environmental dynamism can aid organisations to boost their GOIE. Our key findings are as follows: (a) though green task motivation shows a stronger association with green creativity skills, LSC has a large influence on green creativity skills; (b) green creativity skills exert a strong influence on GOIE while also playing a competitive and complimentary mediating role in our model and (c) environmental dynamism is negatively associated with green creativity skills and GOIE. Furthermore, to validate indirect (v) effects size in *mediation* analysis, we propose a new and more approachable benchmark for v effect size estimations. Organisational and environmental policy implications are discussed.

KEYWORDS

environmental dynamism, environmental policy, environmental sustainability, green creativity, green expertise and motivation, stakeholder engagement, STARA, sustainable development

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1 | INTRODUCTION

With increasing calls to combat global warming and climate change concerns by leveraging radical technological advancements, organisations worldwide are pressured continuously to increase their green creativity (L. Zhu & Zhang, 2019) and innovativeness (Rahman et al., 2020). To do this, organisational team leaders must be competent in managing and deploying *smart technologies*, *artificial intelligence*, *robotics and algorithms* (STARA) that are fundamental to defined green initiatives (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020; Oosthuizen, 2019). Team leaders who are STARA competent can play essential roles in further reducing cumbersome project complexity, enhancing work operations and driving their teams' productivity (Brougham & Haar, 2018). By demonstrating Leader STARA competence (LSC), leaders can adequately support respective teams' efforts and consequently share their STARA related knowledge with their teams. Likewise, LSC can also drive the increased implementation of green creativity-related initiatives relevant to achieving green innovations (Ogbeibu, Burgess, et al., 2020; Oosthuizen, 2019). LSC becomes even more relevant to efficiently drive subsequent achievement of set organisational green innovation objectives (Ogbeibu, Emelifeonwu, et al., 2020; Wong, 2012). Cultivating green creativity (which consists of green task motivation, creativity skills and expertise) and LSC can foster long-term organisational survival, especially given unpredictable environmental (market and geological) conditions (T.M. Amabile, 1997; Frey & Osborne, 2017; Ogbeibu, Emelifeonwu, et al., 2020). According to Ogbeibu, Emelifeonwu, et al. (2020), green creativity is the formation and development of environmentally sustainable and innovative ideas. Similarly, *LSC* is the ability of a leader to demonstrate expertise in STARA in order to more effectively and efficiently drive organisational objectives fundamental to defined tasks (Brougham & Haar, 2018; Oosthuizen, 2019). Green creativity and LSC also help organisations to catalyse green initiatives that bolster green organisational innovative evidence (GOIE) (Jia et al., 2018; Khallash & Kruse, 2012). As an evidence-based strategy, GOIE attracts investors who are keen on advancing sustainable development goals (SDGs) (Sachs et al., 2019).

Though green creativity and LSC could be advocated for curbing global warming concerns, an inadequate effort has so far been directed towards this endeavour (Huang et al., 2014; Ogbeibu, Burgess, et al., 2020). Existing research provides evidence that suggests a growing lack of studies that continues to overlook the plausible multidimensional roles of green creativity and how they respectively influence GOIE (Jia et al., 2018; Li et al., 2020). There is but scant research that deepens insights into how green creativity skill is influenced by other components within the green creativity construct (Amabile & Mueller, 2008; Mittal & Dhar, 2016). Congruent with the seminal work of Amabile (1997), prior studies have continued to overlook the possible mediating roles of green creativity skills and how it influences the relationships between other green creativity components and GOIE (Jia et al., 2018; Li et al., 2020; Song & Yu, 2017). Although recent research investigates green creativity as a unidimensional construct (Li et al., 2020; Ogbeibu, Emelifeonwu, et al., 2020), we argue that this practice suppresses meaningful information that could have otherwise yielded substantive insights for more relevant policy implications. Likewise, as an emerging concept, LSC is yet underdeveloped and remains sparse in the extant literature, and its role on green creativity skills or GOIE still lacks adequate attention (Brougham & Haar, 2018; Khallash & Kruse, 2012; Oosthuizen, 2019). Moreover, when examining human behaviour within organisations, from an environmental sustainability context, studies contend for the need for actors that mirror the environment—such as environmental dynamism—to be considered (Jansen et al., 2009; Ogbeibu, Emelifeonwu, et al., 2020). Nevertheless, the possible influence of environmental dynamism on green creativity skills and GOIE is a crucial factor that present related research and others continue to ignore (Ali et al., 2020; Cop et al., 2020). Consequently, we aim to simultaneously examine the distinct roles of LSC, green creativity and environmental dynamism on GOIE and how they respectively aid organisations to bolster green innovative evidence fundamental to advancing the achievement of the UN's SDGs.

Jansen et al. (2009) and Ogbeibu, Emelifeonwu, et al. (2020) define environmental dynamism as the capacity of the environment to provoke technology variations, oscillations in materials supply and demands of products and customer inclination inequalities. Though the concept of green innovation's influence on the environment has received ample attention in recent years, little is known about how environmental dynamism impacts GOIE (Jansen et al., 2009; Khan & Mir, 2019; Wong, 2012). Among other subsequent contributions, we aim to close these identified gaps in the literature. Although several manufacturing organisations across developed and emerging economies exhibit the potential to produce meaningful green innovations, such potential alone is no guarantee for investments by potential stakeholders (Abdu & Jibir, 2018; Dangelico et al., 2016; Melander, 2017). Given the rising global warming uncertainties and radical technological advancements, organisations consequently need to cultivate their LSC and strengthen their green creativity to bolster further green initiatives that provide evidence for investors (Jabbour, 2013; Xie & Zhu, 2020). However, several organisations that exhibit green creativity struggle to gain or sustain the competitive edge needed to drive their GOIE (Liao & Tsai, 2018; Song & Yu, 2017). Studies postulate that this is partly due to a lack of apt consideration for green creativity components—green task motivation, green expertise and green creativity skills (Amabile, 1997; Song & Yu, 2017). The drive to demonstrate GOIE is a challenge, particularly when organisations and investors are faced by a future replete with constant technological advancements and a volatile environment (Islam et al., 2018; Jung et al., 2018; Zhou et al., 2018).

We attempt to make a fourfold contribution to the green creativity, green innovation and environmental sustainability paradigms. First, by leveraging the long-overlooked components of green creativity, we attempt to deepen insights into how each green creativity component influences GOIE. Second, given recent global warming concerns and a need to combat them by using radical technological advancements, we introduce and empirically examine LSC, an emerging and timely yet much-ignored factor that can act to engender green

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creativity skills, and consequently GOIE. We investigate how LSC, green task motivation and green expertise influences green creativity skills to provoke new insights. Third, we take into consideration plausible external influences of environmental dynamism on green creativity skills and GOIE. Fourth, we investigate the potential distinct mediating roles of green creativity skills on the relationships between green task motivation, green expertise, LSC, environmental dynamism and GOIE. Before reviewing relevant literature, we recognise that we have introduced several constructs, which may confuse the reader. To help alleviate this concern, we provide a list of more simplified definitions of our examined constructs in Table 1. We do not include common acronyms from statistics or vernacular, such as AVE for average variance extracted or HQ for headquarters.

2 | LITERATURE REVIEW

The *GOIE* concept is defined as an organisation's demonstration of green patent creation and applications, quality standards of green innovations, justification of green innovation impact via scientific or

TABLE 1 Alphabetized constructs, their lesser-known acronyms and their simplified descriptions

Long form	Acronym	Simplified definition
Componential theory of organisational creativity and innovation	CTOCI	This theory partly undergirds the concepts of green organisational innovative evidence and green creativity which includes green creativity skills, green expertise and green task motivation (T.M. Amabile, 1997).
Environmental dynamism		The capability of the environment to provoke technology variations, oscillations in materials supply and demands of products, and customer inclination inequalities (Jansen et al., 2009; Ogbeibu, Emelifeonwu, et al., 2020).
Green organisational innovative evidence	GOIE	Tangible products (evidence) of green initiatives (Abdu & Jibir, 2018; Dai & Liu, 2018; Islam et al., 2018; Park et al., 2018).
Green creativity		The ability to create environmentally sustainable novel ideas (Mittal & Dhar, 2016; Ogbeibu, Emelifeonwu, et al., 2020).
Green creativity skills		Green intellectual competence that is relevant for exploring novel pathways for generating green creative ideas to solve problems associated with green initiatives (Mack & Landau, 2015; Ogbeibu et al., 2020; Zabelina & Silvia, 2020)
Green expertise		The combination of green factual knowledge, technical aptitudes, and talents of a team (Chang & Chang, 2017; Tabesh et al., 2019)
Green task motivation		A strong interest inspired by strong challenge or satisfaction, curiosity or intention to accomplish fixed goals for green oriented tasks (Amabile, 1997; Birdi et al., 2016).
Leader STARA competence	LSC	The ability of a leader to leverage STARA competencies to drive organisational objectives (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020; Oosthuizen, 2019)
Smart technologies, artificial intelligence, robotics and algorithms	STARA	A set of contemporary tech-related competencies introduced by Brougham and Haar (2018)

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technical publications and high green technology exports in order to advance the organisation's societal recognition in its innovativeness, increase R&D investments and sustainable environmental impacts (Abdu & Jibir, 2018; Dai et al., 2018; Islam et al., 2018; Park et al., 2018; Wicki & Hansen, 2019). GOIE is important for building integrity and trust with immediate stakeholders and potential investors, especially in times of constant technological changes and environmental volatility (Ardito et al., 2018; Ogbeibu, Emelifeonwu, et al., 2020). This trust has the potential to increase organisational product-market performance (Rahman et al., 2020). GOIE acts to reassure stakeholders and investors that organisations can excel at green innovation (Belderbos et al., 2018; Castellani & Fassio, 2018; Jung et al., 2018). Its entrepreneurship in this respect can also lead to greater company success (e.g., Gali et al., 2020).

Insights into the GOIE concept originate from Ogbeibu, Abdelhak, Tan, et al.'s (2018) study. In their qualitative work, Ogbeibu, Abdelhak, Tan, et al. (2018) argue that a manager's innovative evidence is often a consequence of their reputation for demonstrating their support for the diffusion of creative ideas. Although this finding is meaningful, it mainly captures insights specifically into *managers*' innovative evidence rather than GOIE. Extant research indicates that green creativity could help organisations survive the blows of environmental dynamism and further bolster green innovations (Gilson & Litchfield, 2017; Song & Yu, 2017). Jia et al. (2018) argue that green creativity is becoming more relevant for organisations' increased productivity and competitiveness. Congruently, tenets of green task motivation, expertise and creativity skills have been advocated by T.M. Amabile (1997) and Mittal and Dhar (2016) as significant drivers of green creativity and, without which, team members' creativity may be stifled. Debates of Mittal and Dhar (2016) and Awan et al. (2019) also support that green creativity is a necessary precursor and catalyst of GOIE.

Moreover, several scholarly works have attempted to pursue GOIE by deepening insights into the nexus of green creativity and green innovation (e.g., Awan et al., 2019; Ogbeibu, Emelifeonwu, et al., 2020). Prior research has also suggested positive links between green creativity and related concepts typifying GOIE (Gilson & Litchfield, 2017; Jia et al., 2018; Kawai et al., 2018). However, the significance of GOIE specifically, and the need to further empirically investigate it, have long been overlooked. Despite the proliferation of information on green creativity across diverse cultural contexts worldwide, the influence of green creativity on GOIE, especially within the context of the Nigerian manufacturing industry, is yet to be given empirical consideration (Ogbeibu, Abdelhak, Gaskin, et al., 2018). There is also scant literature that relates implications organisations ought to employ to foster their GOIE from perspectives of the green creativity components, environmental dynamism and LSC collectively (Brougham & Haar, 2018; Ogbeibu, Emelifeonwu, et al., 2020).

Compared to the Nigerian manufacturing industry, several organisations from other developing economies worldwide have begun exemplifying more substantial green creativity via a global recognition of their GOIE (Nasurdin et al., 2014; Sachs et al., 2019). This claim reflects the Nigerian manufacturing industry's lower green innovation position as supported by reports from prior research (Ogbeibu, Abdelhak, Gaskin, et al., 2018; Sanni, 2019). Despite enormous efforts appended by several manufacturing organisations in Nigeria to engender green creativity, and subsequently to foster GOIE, reports show that Nigeria is yet globally less recognised in green creativity and innovation (Ogbeibu et al., 2021; Orji et al., 2019; Sanni, 2019). The Nigerian manufacturing industry is believed to possess the capabilities to position Nigeria's GOIE on the global market, ensuring Nigeria is among the world's top competitive economies in terms of green creativity and GOIE. Excellence in green activities can enhance a country's economic growth (Fernandes et al., 2021). Nevertheless, Nigeria is argued to be grossly underperforming in recent years despite the efforts of its manufacturing organisations (Atanda & Olukoya, 2019; Ogbeibu et al., 2021; Sanni, 2019). Consequently, Nigeria provides a useful case for understanding the gap between green innovative potential and green innovative performance, prevalent in similar developing economies (Ogbeibu et al., 2021; Sachs et al., 2019). By understanding the green creativity drivers of GOIE in Nigeria, we will better understand how to close that gap in similar developing economies and consequently provoke an increase in investors who are willing to support environmental sustainability.

3 | THEORETICAL FRAMEWORK AND HYPOTHESIS DEVELOPMENT

The Amabile's (1997) componential theory of organisational creativity and innovation (CTOCI) and stakeholder theory (Clarkson, 1995) are employed in this study. Amabile and Mueller (2008) suggest that green innovative inputs are a function of green expertise, creativity skills and task motivation. Thus, the extent to which team members can provide evidence of their green innovativeness is contingent on the level of influence they are able to utilise from their respective green expertise, green creativity skills and green task motivation (Birdi et al., 2016; Song & Yu, 2017). *Green expertise* is the combination of green factual knowledge, technical aptitudes and talents of a team, whose objectives and goals are centred on driving green initiatives across disparate green task domains (Chang & Chang, 2017; Tabesh et al., 2019). *Green creativity skills* represent a team's green intellectual competence relevant for exploring novel pathways for generating green creative ideas to solve problems associated with green initiatives (Mack & Landau, 2015; Ogbeibu, Emelifeonwu, et al., 2020; Zabelina & Silvia, 2020). Green task motivation encompasses both intrinsic and extrinsic task motivation (Amabile, 1997). In this study, green task motivation is defined as a strong interest inspired by strong challenge or satisfaction, curiosity or intention to accomplish fixed goals for green-oriented tasks (Birdi et al., 2016; Song & Yu, 2017; Zhu et al., 2016). We acknowledge that various motivation theories have been propounded primarily to examine motivation in prior research (e.g., Ryan & Deci, 2000). However, given the complexity of our model, a further examination of a distinct motivation-related theory would mainly support a minor fraction—green task motivation—of our model. Moreover, the CTOCI propounded by

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Amabile (1997) and further expounded in Amabile and Pratt (2016) provides an overarching theoretical undergirding that incorporates the motivation concept in our study, along with all other constructs examined in our complex model. Therefore, our study's concept of motivation is examined mainly from the theoretical lens of the CTOCI.

Over the years, CTOCI has played several roles to advance the green creativity concept (Mittal & Dhar, 2016; Ogbeibu, Abdelhak, Gaskin, et al., 2018). However, despite the plethora of information underpinning the green creativity concept, extant studies (Mack & Landau, 2015; Tabesh et al., 2019) have overlooked the plausible influences of green expertise and green task motivation on green creativity skills. Given the work of Teece et al. (1997), green creativity skills implicitly mirror a difficult-to-trade knowledge asset by which an organisation may reconfigure, transform and further bolster its dynamic capabilities fundamental to fostering GOIE. As a probable driver of an organisation's dynamic capabilities needed to advance their GOIE, the works of Teece et al. (1997) and Amabile (1997) relate that green creativity skills can help organisations better respond to the volatility of environmental impacts. However, despite the importance of extant research exploring green creativity skills, studies (Mack & Landau, 2015; J. Wang et al., 2017) continue to ignore the possibility of theorising a causal inference between the green creativity components advocated in the CTOCI. Congruent with Amabile (1996), CTOCI overlooked the plausible relationship between green expertise and green creativity skills and how they influence GOIE. Equally, given recent debates on changes in technological advances, there is also a need to deepen insights into the probable influence of LSC on green creativity skills, and consequently on GOIE, as this was not empirically considered by the CTOCI (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020). This exemption might have been due to the gap in time between recent STARA impacts and the time CTOCI was introduced (Amabile, 1996, 1997; Brougham & Haar, 2018). Although CTOCI advocated the need not to overlook the impact of environmental dynamism on the creative process, much has been left unknown as to how environmental dynamism influences green creativity skills and GOIE, respectively (Huang et al., 2014; Jansen et al., 2009). It is likewise unclear how green creativity skills act to mediate the relationships between green expertise, LSC, green task motivation and environmental dynamism on GOIE (Zabelina & Silvia, 2020). These theoretical gaps (that our study aims to fill) strengthen the need to advance the CTOCI further to ensure consistent and timely relevance and relayed meaningful implications on environmental sustainability (Jung et al., 2018).

Moreover, although the CTOCI advocates meaningful implications for organisations within a defined environment, it does not yet inform how organisations ought to respond to the call to foster environmental sustainability (Yong et al., 2019). Thus, the gap between profiteering and bolstering of environmental sustainability demands closer attention. Freeman et al. (2018) argue for organisations to adopt stakeholder theory to bridge this gap. Stakeholder theory advocates an interface between organisations and the wider environment (Clarkson, 1995; Ranangen, 2015). According to Barney and Harrison (2020), stakeholder theory accentuates the need for organisations to not mainly focus on profiteering, but on the environment from which they extract their resources. Yong et al. (2019) lament that ample care should be directed towards global warming concerns, which is a consequence of innovations that negatively influence the environment. Although CTOCI contributes to driving green creativity tenets towards engendering GOIE—consequently maximising profits—stakeholder theory aids to further ensure organisations' profit maximising objectives are consistently aligned with environmental sustainability demands (Ogbeibu, Emelifeonwu, et al., 2020).

3.1 | Effects of green expertise and green task motivation on green creativity skills

Green expertise and green task motivation are two core drivers of green creativity (Chang & Chang, 2017; Li et al., 2020). CTOCI suggests that team members with relevant green expertise and motivation tend to be able to demonstrate resilience and prove resourceful in their defined green initiatives and innovation task domains (Amabile & Mueller, 2008; Cop et al., 2020; Junsheng et al., 2020). Green expertise and green task motivation are probable drivers organisations could leverage to foster green creativity skills (Ali et al., 2020; Chang & Chang, 2017). Similarly, green task-motivated experts are usually a necessary catalyst for green creativity skills because in addition to being motivated (Ali et al., 2020). They also normally have quick access to relevant information from prior green-related experiences (Dayan et al., 2013; Delgado, 2010). Extant research argues that having green expertise and being green task-motivated could support team members' demonstration of adeptness, flexibility and perseverance, which are important qualities needed to drive green initiatives (Delgado, 2010; Liang et al., 2018). Green expertise and green task motivation are thus needed to drive the creative thinking process evidenced by exhibited green creativity skills to provoke green solutions that are fundamental to bolstering environmental sustainability (Ali et al., 2020; Chang & Chang, 2017).

Hypothesis 1. Green expertise has a positive association with green creativity skills.

Hypothesis 2. Green task motivation has a positive association with green creativity skills.

3.2 | The effects of LSC on green creativity skills

The debate undergirding LSC is still in its embryonic phase, and research that has delved into the possible association of LSC and green creativity skills lacks empirical attention (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020). To close this gap and subsequently foster meaningful policy implications, there is a need to deepen insights into how LSC influences green creativity skills (Oosthuizen, 2019). The works of Brougham and Haar (2018) and

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Oosthuizen (2019) suggest that with growing technological advancements in STARA, the future of work faces a dire uncertainty that if overlooked by organisational leaders could subsequently render some operations obsolete. Frey and Osborne (2017) and Makridakis (2017) contend that organisations ought to consider cultivating timely competencies, especially those fundamental to STARA. Castellani and Fassio (2018) and Ogbeibu, Burgess, et al. (2020) support that LSC is relevant for further improving work operations, reducing cumbersome project complexity and driving productivity. Moreover, given the drive to foster environmental sustainability, extant research contends that STARA competencies could foster speed and efficiency and save time allocated for driving green initiatives (Vishwanath et al., 2019). Knowledge of STARA's myriad benefits is becoming more prevalent, and organisational leaders are consequently pressured to adopt, adapt and cultivate STARA-related competencies (Berrone et al., 2013; Parker & Grote, 2020). By introducing LSC into the work processes, work teams can further explore, exploit and garner novel thinking pathways by which green tasks can be initiated and completed (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020). Consequently, by leveraging STARA competencies, leaders are able to support their teams' efforts further and drive increased application of creative thinking skills needed for achieving green expectations (Frey & Osborne, 2017; Vishwanath et al., 2019).

Hypothesis 3. LSC has a positive association with green creativity skills.

3.3 | The effects of environmental dynamism on green creativity skills and GOIE

An organisation's external environment, such as customers, technological advancements, competitors, governmental institutions and suppliers within disparate business systems (Chan et al., 2016), impacts how the organisation reacts towards the production of green innovations (Ogbeibu, Emelifeonwu, et al., 2020). Because the greening of innovations requires the exhibition of green creativity skills by team members who are often engaged in the production and implementation of green creative ideas, studies advocate that environmental dynamism can cause organisations to mount increasing pressure on team members to exert higher green creativity skills to demonstrate GOIE (Kawai et al., 2018; Ogbeibu, Emelifeonwu, et al., 2020). Though significant associations have been noted between environmental dynamism and GOIE, how environmental dynamism influences teams' green creativity skills, and GOIE, simultaneously, lacks empirical attention (Chan et al., 2016; Ogbeibu, Emelifeonwu, et al., 2020). Congruent with the volatility of environmental impacts on business operations, organisations are compelled to re-evaluate their business innovation models underpinning green initiatives (Huang et al., 2014). Studies lament that this could add challenges to work teams already pressured to produce and exhibit green creativity skills needed to combat environmental dynamism's unpredictable impacts on green initiatives (Berrone et al., 2013; Kawai et al., 2018). Such constant pressure on work teams due to environmental dynamism could increase stress, provoke work-life balance conflict, and hamper GOIE as team members struggle to meet constantly evolving changes and demands of environmental actors (Khallash & Kruse, 2012; Ogbeibu, Emelifeonwu, et al., 2020). By the time team members complete the ideation and successful implementation of a particular green creative idea amid dynamic environmental influences, the GOIE might no more be of particular interest to stakeholders who, by then, may have shifted focus toward newer disruptive technologies (Chan et al., 2016; Ogbeibu, Emelifeonwu, et al., 2020).

Hypothesis 4. Environmental dynamism has a negative association with green creativity skills.

Hypothesis 5. Environmental dynamism has a negative association with GOIE.

3.4 | Green expertise and GOIE

Green expertise, as an accumulation of green work experience (Pandita et al., 2019) is an important component that can support the application of green creativity and subsequent GOIE (Gube & Lajoie, 2020). Prior research expounds that Green expertise is the basis of all green innovation tasks and is relevant for addressing complex and technical issues regardless of the green task domain they are identified with (Boshuizen et al., 2020; Howell, 2017). Team members who have proficient knowledge regarding a defined green innovation initiative can help their organisations evidence innovativeness (Dayan et al., 2013). Tang et al. (2014) and Delgado (2010) argued that the exhibition of green expertise is often a function of prior green knowledge or direct experience deployed to drive objectives for fostering the successful completion of set green innovation initiatives. Green expertise can expedite further the accomplishment of green innovation initiatives by which organisations may demonstrate their GOIE (Boshuizen et al., 2020). Moreover, GOIE is an investment boosting strategy forged out from a successful integration and coordination of team members' green expertise (Chang & Chang, 2017; Jia et al., 2018). Debates expounding on CTOCI further suggest that team members may often feel pulled to exert more green innovative behaviours and produce meaningful GOIE should they possess the specific required expertise (Amabile & Mueller, 2008; Chang & Chang, 2017). Consequently, such team members may further boost their organisation's green innovative evidence (Tang et al., 2014).

Hypothesis 6. Green expertise has a positive association with GOIE.

3.5 | Green creativity skills and GOIE

Green creativity skills deal with how team members think and divergently process green ideas to produce novel solutions for environmental sustainability concerns (Ali et al., 2020; Amabile &

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Mueller, 2008). Studies also debate that green creativity skills has a positive association with GOIE (Baer, 2012; Ogbeibu, Emelifeonwu, et al., 2020). By exploring several distinct and innovative ways of identifying green problems and proffering practical green solutions, team members can provide meaningful green innovations by which their organisation may further exemplify GOIE (Li et al., 2020). Leaders who focus on nurturing team members' green creativity skills tend to benefit from having a workforce that is adept at thinking divergently, thus boosting their organisation's workforce innovativeness (Gube & Lajoie, 2020; Song & Yu, 2017). Therefore, harnessing green creativity skills is important for generating green creative ideas which an organisation needs to bolster its competitive edge and GOIE (Awan et al., 2019; Sanni, 2019).

Hypothesis 7. Green creativity skills has a positive association with GOIE.

3.6 | Green task motivation and GOIE

Green task-motivated team members often demonstrate risk-taking behaviours, flexibility, commitment, perseverance and willingness to drive themselves towards achieving GOIE-related objectives (Ali et al., 2020; Dewett, 2007). Green task motivation (mainly green intrinsic motivation) is the principal driver of creativity (Amabile, 1997; Junsheng et al., 2020). Liang et al. (2018) and Li et al. (2020) emphasise that green task motivation reflects team members' inner needs that result in gratification received for increases in their green creativity and demonstration of innovativeness. Junsheng et al. (2020) also suggest that green task motivation and the zeal for organisations to exemplify their innovativeness is usually triggered by team members' green task motivation. Zhu et al. (2016) relate that green task-motivated team members are likely to show interests, curiosity, and satisfaction towards creative initiatives like problem identification, creative ideation and implementation. Green task motivation has been theorised to drive GOIE in the course of organisations' pursuits for socio-economic impact and global recognition for innovativeness (Li et al., 2020). Therefore, green task-motivated team members are relevant for reinforcing defined objectives needed by organisations to bolster their GOIE (Ali et al., 2020; Awan et al., 2019).

Hypothesis 8. Green task motivation has a positive association with GOIE.

3.7 | The mediating role of green creativity skills between green expertise and GOIE

According to Sasidharan (2018), team members highly proficient at executing specific domain tasks are often more likely to demonstrate higher innovativeness when dealing with challenges arising from their areas of expertise. Tang and Naumann (2016) and Chang and Chang (2017) stress that green expertise positively influences GOIE. The work of Wang et al. (2017) infers that green expertise is a prime factor required to engender GOIE. With the required green expertise, team members might exert relevant knowledge to solve a domain-specific green task challenge (Sasidharan, 2018). However, by employing green creativity skills, team members may efficiently utilise prior green expertise in distinct ways to address a specific complex challenge, effectively manage projects and save time and resources (Gube & Lajoie, 2020; Wang et al., 2019). In this regard, green creativity skills could help team members amplify their respective green expertise implementation via extensive exploration of several novel cognitive pathways stemming from prior experience or knowledge (Gube & Lajoie, 2020). Therefore, green creativity skills are relevant for helping experienced team members further drive green initiatives that are fundamental to engendering GOIE (Baer, 2012; Ogbeibu, Emelifeonwu, et al., 2020). Green creativity skills consequently provide a positive mediating influence in that it can transfer the positive impact of team members' expertise on GOIE (Guo & Tang, 2017).

Hypothesis 9. Green creativity skills mediate the positive association between green expertise and GOIE.

3.8 | The mediating role of green creativity skills between green task motivation and GOIE

In the course of executing challenging green innovation-related initiatives, green task-motivated team members tend to exhibit resilience, commitment, perseverance and willingness (Dewett, 2007; Junsheng et al., 2020). However, being task-motivated may not necessarily guarantee team members' efficient identification of ill-defined problems, rapid generation of creative ideas or successful implementation of green innovations (Davis et al., 2020; Li et al., 2020). Prior research advocates team members' need to leverage a more divergent cognitive approach (i.e., creativity skills) to address practical problems by producing and exploring novel solutions (Runco, 2013; Song & Yu, 2017). Thus, green task-motivated employees tend to be more open to thoughtful exploration to produce creative ideas that could further bolster GOIE (Abdu & Jibir, 2018; Dewett, 2007). Recent research accentuates that green creativity skills present more meaningful results for team members who, being task motivated, desire to achieve innovative outcomes relevant for engendering GOIE (Li et al., 2020; Liang et al., 2018).

Hypothesis 10. Green creativity skills mediate the positive association between green task motivation and GOIE.

3.9 | The mediating role of green creativity skills between LSC and GOIE

Concepts associated with green creativity skills have been examined in prior research (Jung et al., 2018; Song & Yu, 2017). However, due

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to distinct contextual and empirical approaches employed in extant research, findings obtained from relative debates of green creativity skills are inconclusive (Gube & Lajoie, 2020; Mittal & Dhar, 2016). In an effort to effectively utilise STARA competencies amid constant technological advancements, Ogbeibu, Burgess, et al. (2020) argue that green creativity skills is a core factor that can help leaders to catalyse the implementation of green objectives further. Green creativity skills are a set of intellectual capabilities that foster productivity in work processes and reduce work complexities (Gube & Lajoie, 2020). Green creativity skills provide an important key by which leaders can explore more divergent ways of leveraging their STARA competencies to solve green issues impeding the achievement of GOIE (Awan et al., 2019; Brougham & Haar, 2018). Congruent with the debates of Amabile (1997) and Frey and Osborne (2017), green creativity skills deal with providing that 'something extra' that can provoke leaders to perform more creatively while implementing STARA competencies to engender GOIE further. However, despite extant literature support, much is yet to be done to deepen insights that empirically establish the alleged role of green creativity skills in the relationship between LSC and GOIE (Awan et al., 2019; Brougham & Haar, 2018).

Hypothesis 11. Green creativity skills mediate the positive association between LSC and GOIE.

3.10 | The mediating role of green creativity skills between environmental dynamism and GOIE

In the face of uncertainty fuelled by the influence of environmental dynamism on organisations, Li et al. (2020) and Ogbeibu, Emelifeonwu, et al. (2020) contend that organisational workforces need to exhibit increased green creativity behaviours to engender and sustain GOIE. Gube and Lajoie (2017) elucidate that by constantly tapping into and exploiting an organisation's green creativity skills, an organisation might survive the blows of a constantly evolving business environment. Moreover, although organisational demand for increased green creativity skills might cause increased pressure on team members, team members may be provoked to exhibit green creativity skills to foster GOIE (Berrone et al., 2013; Junsheng et al., 2020; Song & Yu, 2017). Consequently, increased green creativity skills can act as a plausible dampener to the otherwise negative influence of environmental dynamism on GOIE (Ogbeibu, Emelifeonwu, et al., 2020). By constantly deploying green creativity skills to explore and exploit novel pathways to combat the probable varying effects of environmental dynamism on GOIE, team members may be able to predict, prepare against and foster organisational resilience against environmental dynamism influences (Huang et al., 2014; Jansen et al., 2009). The recent debate suggests that, although environmental dynamism has a negative influence on GOIE, the presence of green creativity skills is a key factor for sustaining underlying green business processes and related initiatives fundamental to GOIE (Ogbeibu, Emelifeonwu, et al., 2020).

Hypothesis 12. Green creativity skills mediate the negative association between environmental dynamism and GOIE.

The hypotheses are presented diagrammatically in Figure 1.

4 | METHODOLOGY

4.1 | Sample size and data collection procedure

This study's target population is represented by a sample of 341 team members plus 52 team leaders, comprising 393 respondents. These respondents belong to teams from the research and development (R&D) and Information Technology (IT) departments of 55 different manufacturing organisations' headquarters (HQ) located in Lagos, Abuja and Port Harcourt of Nigeria. All 55 organisations were medium-size enterprises with employee counts ranging from 75 to 200. Data collection was obtained from 122 participants from 18 manufacturing organisations in Port Harcourt, from 89 participants from 12 manufacturing organisations in Abuja and 182 participants from 25 manufacturing organisations in Lagos, respectively. The teams ranged from four to eight employees. As advocated by extant research, the chosen locations represent hubs of manufacturing organisations in Nigeria (Ogbeibu, Abdelhak, Gaskin, et al., 2018). This study used the Krejcie and Morgan (1970) sample size determinant to obtain a stratified proportionate sampling of participants. Although 488 questionnaires were distributed, 393 completed questionnaires were returned and found useful for subsequent analysis, thus indicating an 80% response rate. This result is consistent with prior related research (Ogbeibu, Abdelhak, Gaskin, et al., 2018). Respondents were between 20 to 60 years old. Concerning gender, 56.6% were male and are thus not overrepresented in this study. Additionally, 4% of respondents had a diploma or equivalent, 52% had undergraduate degrees, 40% had a master's degree and only 4% had a PhD.

The questionnaire was prepared in English and comprised 7-point Likert agreement scales and demographic questions. Trained research assistants (RAs) were hired to liaise with each manufacturing organisation's human resource manager (HRM) to execute this study's data collection process. This study acknowledges the contentions regarding the use of cross-sectional design (Rindfleisch et al., 2008). Consequently, as Ogbeibu, Emelifeonwu, et al. (2020) recommended, we employed a temporal separation (time lag) between the periods of data collection of the endogenous and exogenous constructs of our study. Therefore, questionnaires for GOIE and environmental dynamism were distributed 9 weeks after LSC distribution, green task motivation, green creativity skills and green expertise. This approach supported measures taken to pre-empt common method bias (CMB) and common source bias (CSB) (Kock, 2015; Podsakoff et al., 2012; Roese & Vohs, 2012). Likewise, participants were ensured of their anonymous participation, and an item from LSC, green task motivation and environmental dynamism measures was respectively reverse coded to catch unengaged respondents (Podsakoff et al., 2012).

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Additionally, consistent with Kock's (2015) recommendation for addressing CMB, Table 2 of this study highlights that the highest variance inflation factor (VIF; 2.831) is less than the threshold of 3.3. Therefore, it can be inferred that CMB did not significantly influence participants' responses. These study's data have come from multiple sources, thus curbing CSB influence as participants' responses originated from team leaders and team members. Team leaders assessed their respective team members and themselves by answering the questions on green expertise and green creativity skills. Team members accessed their leaders by answering measures of LSC. Moreover, measures of green task motivation, environmental dynamism and GOIE were answered by team members and their leaders. Green task motivation has been assessed via self-report scales because leaders and team members were required to tap into what truly motivated them intrinsically. It is unlikely that team leaders' examination of their team members' green task motivation would yield accurate, objective answers that reflect what motivates their team members (Abbas & North, 2018; Dewett, 2007).

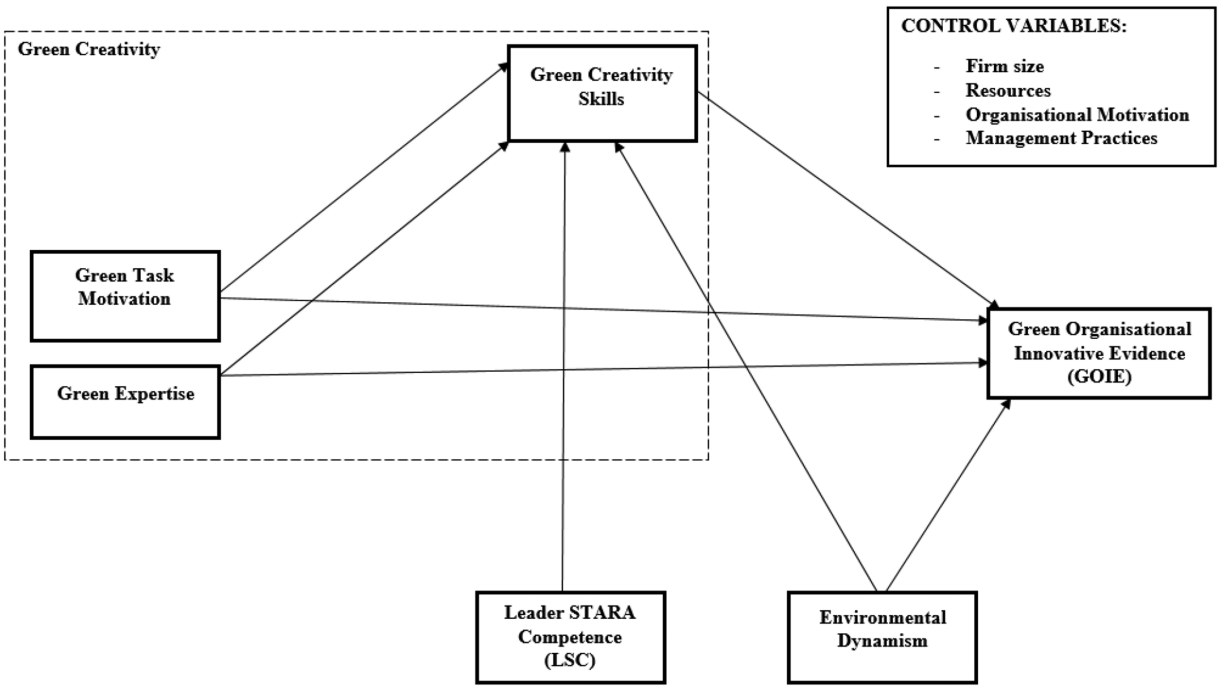

FIGURE 1 Theoretical model

TABLE 2 Reliability, validity and overall model fit assessment based on PLS-SEM results

Construct	rho_A	Composite reliability (CR)	AVE	VIF values		Overall model fit assessment and *P* values at 0.5 confidence intervals in brackets		
				GOIE	GCS	SRMR	d_G	d_{ULS}
Environmental dynamism	0.876	0.868	0.572	1.806	1.698	0.059 (0.035)	1.303 (0.332)	1.513 (0.528)
Firm size	1.000	1.000	1.000	2.421				
Green organisation innovative evidence (GOIE)	0.866	0.866	0.683					
Green creativity skills (GCS)	0.858	0.847	0.584	1.831				
Green expertise	0.955	0.954	0.839	1.517	1.370			
Green task motivation	0.868	0.858	0.550	2.831	2.141			
Leader STARA competence (LSC)	0.848	0.841	0.572		1.144			
Management practices	1.000	1.000	1.000	2.127				
Organisational motivation	1.000	1.000	1.000	2.395				
Resources	1.000	1.000	1.000	2.503				

Abbreviations: AVE, average variance extracted; VIF, variance inflation factor.

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4.2 | Measures

Consistent with the recommendation of Amabile and Mueller (2008, p.38), "the major approaches" of prior research (Birdi et al., 2016; Mack & Landau, 2015) have been employed to measure green creativity components (see Appendix A for the full measurement scales). To measure green task motivation, four items were adapted from Burr and Cordery (2001) and one item from Birdi et al. (2016) (0.79 scale reliability). An example of the items used is "I am strongly motivated by the recognition I can get from my company for engaging in a green initiative." Four items were adapted from Birdi et al. (2016) to measure green creativity skills (0.90 scale reliability). An example of the items is "This team member is skilled at generating more than one solution to a problem associated with green initiatives". To measure green expertise, four items were adapted from Birdi et al. (2016) (0.76 scale reliability). An example of the items is "This team member can address almost any problem in his/her green related tasks." To measure environmental dynamism, five items were adapted from Jansen et al. (2009) (0.90 scale reliability). An example of the items is "In our market, nothing has changed in one year" (reverse coded). Based on the works of Brougham and Haar (2018) and S. Ogbeibu, Burgess, et al. (2020), four items have also been self-developed to measure LSC. An example of the items is "My leader has the knowledge and ability to apply smart (analysing, reporting and self-monitoring systems) technology during operations." Based on extant research (Baer, 2012), and indicators for innovativeness as highlighted in Cornell University, INSEAD, and WIPO (2018), three items have been self-developed to investigate GOIE. These items are "My organisation currently owns one or more patented inventions that can positively contribute to environmental sustainability." Finally, congruent with extant research, firm size, management practices, organisational motivation and resources were controlled for in our study (Amabile, 1997; Ogbeibu, Emelifeonwu, et al., 2020).

4.3 | Analysis

The partial least squares structural equation modelling (PLS-SEM) technique was applied to examine this study's data (for further details on why, when and how to use PLS-SEM, see Lowry & Gaskin, 2014; Sarstedt et al., 2016). To foster analytic rigour and more consistent estimates, we employed the consistent PLS (PLSc) approach via SmartPLS 3 (Henseler & Schuberth, 2020; Sarstedt et al., 2016). Furthermore, PLS-SEM was selected for its model specification, ease of interpretation and soft distributional assumptions (Sarstedt et al., 2016).

5 | EMPIRICAL FINDINGS

As a preliminary step, we first examined for the normality of our data (Hair et al., 2010). Standard deviation results (1.015 to 1.673) suggests an even dispersion. Values for kurtosis and skewness range from −1.2 to 0.331 and −0.349 to 0.005, respectively. This confirms normal data distribution (Hair et al., 2010). We next examine the measurement and structural models.

Figure 2 shows that the measurement model's scale items contribute substantially to their respective constructs, given their respective loadings (Ringle et al., 2018) and further evidenced in their significant t-statistics values highlighted in Figure 3. Following the recommendation of Noor et al. (2018), we retained items that loaded above 0.5. Noor et al. (2018) advocated that scale items that load between 0.5 and 0.7 should be retained, but this should be contingent upon average variance extracted (AVE) and composite reliability (CR), meeting their recommended thresholds of 0.5 and 0.7, respectively. Hair et al. (2010) further support their retention because they do not hamper model integrity.

In Table 2, values of rho_A and CR also confirm all constructs' reliability and internal consistency (Henseler, 2017). Similarly, AVE values indicate convergent validity for this study's constructs (Sarstedt et al., 2017). In order to account for any issues of multicollinearity, VIF values have been examined. VIF values highlighted in Table 2 suggest that these study's data are free of multicollinearity concerns (Kock, 2015). Additionally, the discriminant validity has been examined using the Heterotrait-Monotrait Ratio (HTMT), a higher boundary criterion for discriminant validity assessments in PLS-SEM (Henseler, 2017). Consistent with Sarstedt et al. (2017) recommendations, HTMT results of Table 3 indicate that this study's constructs are distinct from each other, confirming discriminant validity.

To examine the structural model, studies (Ringle et al., 2018; Sarstedt et al., 2017) recommend assessing measures such as the R^2, β, statistical significance, f^2, Q^2 and SRMR. Figure 2 displays the endogenous constructs' R^2 and path coefficients. Regarding R^2 measures, we follow the recommendation by Chin (1998), who advocated that R^2 values of 0.67, 0.33 and 0.19 indicate strong, moderate and weak R^2 values, respectively. In terms of effect sizes (f^2), Ringle et al. (2018) advocate that f^2 scores of 0.02, 0.15 and 0.35 suggest small, medium and large effects. To test for statistical significance, Hair et al. (2011) recommend a minimum t value of 1.65 at $p \leq 0.1$. Additionally, Q^2 scores above zero for a certain endogenous construct indicate an acceptable level of predictive accuracy (Ringle et al., 2018; Sarstedt et al., 2017). Based on the recommended measurement criteria espoused by extant research (Ringle et al., 2018; Sarstedt et al., 2017) for determining models' predictive performance, Q^2 results of creativity skills (0.186) and OIE (0.526) suggest acceptable predictive accuracy for the exogenous constructs.

This study used the consistent PLS bootstrapping option with 5,000 subsamples to estimate the structural model (Lowry & Gaskin, 2014). Congruent with extant research (Hu & Bentler, 1999; Sarstedt et al., 2016), Henseler (2017) advocates that the SRMR is the only approximate model fit criterion for examining PLS path modelling. Additionally, we examine the exact model fit by using the bootstrap-based test to obtain values for the discrepancy measures which include the squared Euclidean Distance (d_{ULS}) and the Geodesic Distance (d_G) (Henseler & Schuberth, 2020). As shown in Table 1, the corresponding values of discrepancy measures, including the SRMR value, indicate a good measurement model fit for this study

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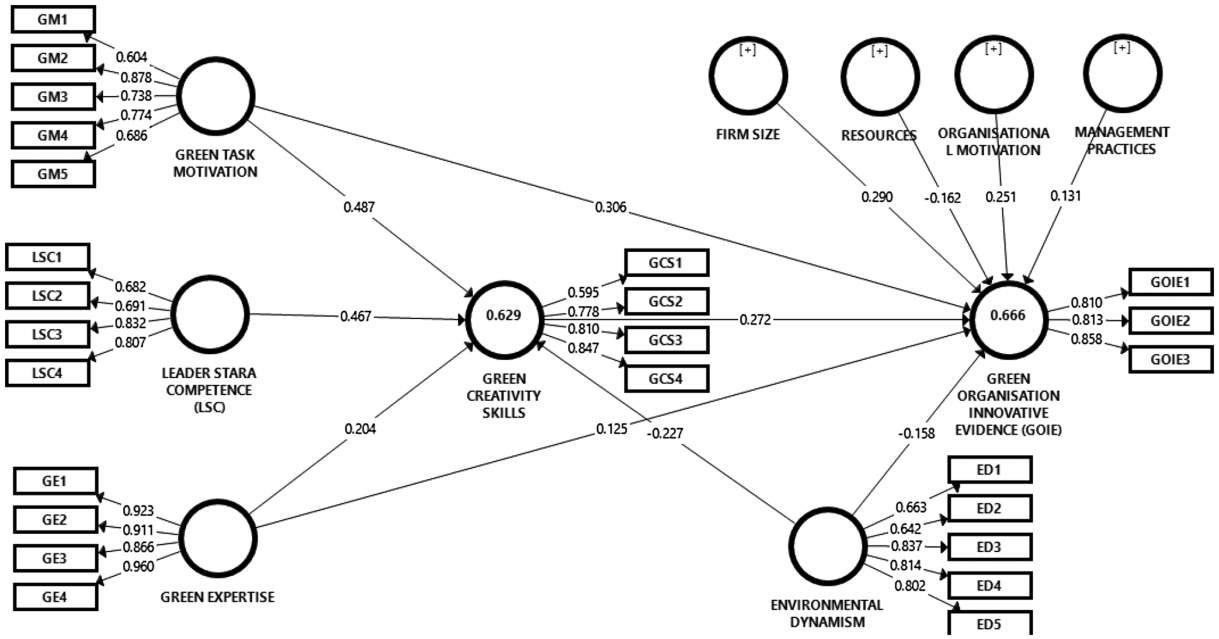

FIGURE 2 SmartPLS3 measurement model

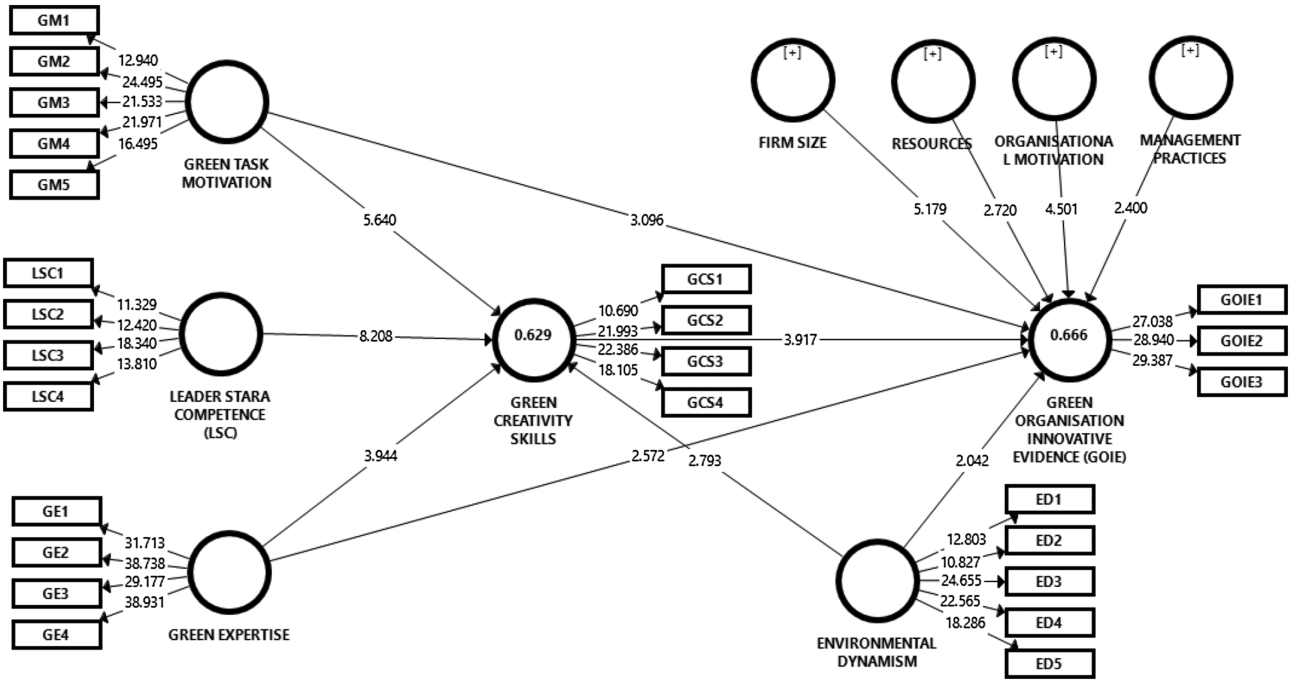

FIGURE 3 SmartPLS3 structural model

and further suggest that our specified model adequately fits our collected data (Henseler & Schuberth, 2020; Hu & Bentler, 1999).

Figure 3 shows that R^2 values of green creativity skills (0.629, t = 11.580, $p \leq .001$) and GOIE (0.666, t = 13.488, $p \leq .001$) indicate significant and relatively strong degree of variance explained. Thus, all together, our exogenous constructs provide significant explanations of variance in green creativity skills and GOIE. Congruent with the respective constructs' path coefficients effect sizes, results of the inner (structural) model in Figures 2 and 3 show that green task motivation has a small (f^2 = 0.099) positive influence on GOIE and a medium (f^2 = 0.299) positive influence on green creativity skills. Although green creativity skills has a positive and relatively medium (f^2 = 0.121) influence on GOIE, green expertise exhibits a small positive influence on green creativity skills (f^2 = 0.082) and GOIE (f^2 = 0.031). Whereas LSC is shown to exhibit a large (f^2 = 0.514) positive influence on green creativity skills, environmental dynamism shows a small negative influence on green creativity skills (f^2 = 0.082) and GOIE (f^2 = 0.042). The above results consequently provide

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support for **H1** to **H8**. Results of the control variables suggest that although firm size exerts a positive and relatively medium (f^2 = 0.104) influence on GOIE, management practices (f^2 = 0.024) and organisational motivation (f^2 = 0.079) each have small positive influences on GOIE. Moreover, resources are shown to have a small (f^2 = 0.032) negative influence on GOIE.

To account for the mediating roles of green creativity skills, we follow Aguinis et al.'s (2016) recommendations and Nitzl et al. (2016). Moreover, to obtain specific indirect (*v*) effect sizes, researchers are advised to square the respective standardized *v* effects (Lachowicz et al., 2018) and then to apply the same benchmark thresholds advocated by Cohen (1988), namely, small >0.02, medium >0.15 and large

TABLE 3 Heterotrait-Monotrait ratio (HTMT)

	ED	FIRM SIZE	GOIE	GCS	GE	GTM	LSC	MP	Org. Mot	Resources
Environmental dynamism (ED)										
Firm size	0.216									
Green organisation innovative evidence (GOIE)	0.262	0.556								
Green creativity skills (GCS)	0.266	0.218	0.591							
Green expertise (GE)	0.289	0.134	0.467	0.507						
Green task motivation (GTM)	0.638	0.338	0.599	0.580	0.500					
Leader STARA competence (LSC)	0.260	0.238	0.382	0.628	0.259	0.322				
Management practices (MP)	0.233	0.593	0.536	0.253	0.237	0.307	0.292			
Organisational motivation (org. Mot)	0.277	0.625	0.570	0.242	0.193	0.339	0.212	0.684		
Resources	0.291	0.716	0.438	0.279	0.141	0.340	0.236	0.576	0.647	

TABLE 4 Summary of constructs' associations and influences

Hypothesis	Constructs	Outcome	Significance	Influence (f^2 — effect size)	Remarks
H1	Green expertise has a positive association with green creativity skills.	Positive association	Significant	Small	Supported
H2	Green task motivation has a positive association with green creativity skills.	Positive association	Significant	Medium	Supported
H3	LSC has a positive association with green creativity skills.	Positive association	Significant	Large	Supported
H4	Environmental dynamism has a negative association with green creativity skills.	Negative association	Significant	Small	Supported
H5	Environmental dynamism has a negative association with GOIE	Negative association	Significant	Small	Supported
H6	Green expertise has a positive association with GOIE.	Positive association	Significant	Small	Supported
H7	Green creativity skills has a positive association with GOIE.	Positive association	Significant	Relatively medium	Supported
H8	Green task motivation has a positive association with GOIE.	Positive association	Significant	Small	Supported
H9	Green creativity skills mediates the positive association between green expertise and GOIE.	Complementary mediation	Significant	<A small effect	Supported
H10	Green creativity skills mediates the positive association between green task motivation and GOIE.	Complementary mediation	Significant	small effect	Supported
H11	Green creativity skills positively mediates the association between LSC and GOIE.	Complementary mediation	Significant	small effect	Supported
H12	Green creativity skills mediates the negative association between environmental dynamism and GOIE	Competitive mediation	Significant	<A small effect	Supported

Abbreviations: LSC, leader STARA competence; GOIE, green organisational innovative evidence.

[Correction added on 6 September 2021, after first online publication: In Table 4, the Influence column for H10 & H11 rows have been corrected to "small effect" in this version.]

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>0.35 However, we advise that *v* effect sizes calculated in this way will rarely reach such thresholds because the *v* effect size is the result of multiplying a decimal by another decimal and then squaring it – in which each step diminishes the product. Thus, to reach a 'small' effect size in this way, the original coefficients of the two constituent paths would be required to average above 0.376 for small, 0.622 for medium and 0.769 for large. Therefore, for *v* effect sizes, **we propose** a halving adjustment to Cohen's recommendations such that the squared standardized *v* effect should be greater than **0.175** for a **large effect**, **0.075** for **medium** and **0.01** for **small**, making them more appropriate for indirect effects.

[Correction added on 6 September 2021, after first online publication: The *v* effect sizes have been corrected in this version.]

Therefore, results for **H9** ($\beta = 0.056$, $t = 2.478$, $p \leq .001$, $v = 0.003$), **H10** ($\beta = 0.133$, $t = 3.378$, $p \leq 0.001$, $v = 0.017$) and **H11** ($\beta = 0.128$, $t = 3.625$, $p \leq 0.001$, $v = 0.016$) show that the positive associations between green expertise, green task motivation, LSC and GOIE are mediated by green creativity skills via 'complimentary mediation' (Nitzl et al., 2016). These results indicate that **H9** to **H11** are supported. Although, concerning the complimentary mediation effect size, green creativity skills would be classified as a 'small' effect for H10 and H11, and less than a 'small' effect for H9. Results for H12 ($\beta = -0.062$, $t = 2.332$, $p \leq 0.020$, $v = 0.003$) also suggests that green creativity skills mediate the negative association between environmental dynamism and GOIE via 'competitive mediation' (Nitzl et al., 2016). Again, given the effect size, green creativity skills would be classified to have less than a 'small' effect in this relationship.[1] Given the nature of this study's model and aims, it is argued that the basic principle of parsimony in the mediation context is upheld (Lachowicz et al., 2018). Consequently, the above results provide support for **H9** to **H12**. Table 4 summarises our findings.

6 | DISCUSSION AND CONCLUSION

Consistent with extant research (Chang & Chang, 2017; Li et al., 2020), our study findings indicate that green expertise and green task motivation have positive associations with green creativity skills. Team members who are motivated to drive green practices and possess adequate green expertise can positively influence the degree to which green creativity skills may be deployed to foster green initiatives, and this is consistent with Stakeholder Theory (Amabile & Mueller, 2008; Junsheng et al., 2020). Though green expertise exhibits a small positive influence on green creativity skills, we find that green task motivation exerts a medium positive influence on green creativity skills. This finding also complements the CTOCI assumption of a stronger positive influence of green task motivation on green creativity skills than green expertise (Amabile, 1996). These conclusions are based on their respective magnitudes of effect sizes. Moreover, we find that LSC exhibits a large positive influence on green creativity skills, consistent with prior research (Brougham & Haar, 2018; Ogbeibu, Burgess, et al., 2020). Our findings relate that environmental dynamism is negatively associated with green creativity skills and GOIE. Though the effect sizes are small, Lowry and Gaskin (2014) advocate that small effects can still contribute significant meaning to our understanding of a phenomenon. Although these findings are congruent with the debate of extant scholarly works (Kawai et al., 2018; Ogbeibu, Emelifeonwu, et al., 2020), it does stand in contrast to the positive notion that the volatile influence of environmental dynamism can provoke teams to increasingly exert green creativity skills to combat pressures from environmental dynamism (Chan et al., 2016; Zhou et al., 2018). Consequently, environmental dynamism hampers green creativity skills and GOIE. Contrary to the postulations of the CTOCI that green task motivation is the strongest driver of green creativity skills, we find that LSC exerts a greater positive influence on green creativity skills compared to other predictors such as green expertise and green task motivation. This inference is based on the magnitude of effect size evidenced in our study. Although Figure 2 shows that green task motivation has the strongest positive association with green creativity skills, the effect size results reveal that LSC exerts the largest positive influence on green creativity skills (Figure 3).

Additionally, we find that though green expertise is positively associated with GOIE, it mainly exerts a small influence on GOIE, consistent with prior research (Boshuizen et al., 2020; Howell, 2017). Likewise, green creativity skills and green task motivation are positively associated with GOIE, and both further exert a relatively medium and small positive influence on GOIE, consistent with extant studies (Ali et al., 2020; Awan et al., 2019; Baer, 2012; Li et al., 2020; Ogbeibu, Emelifeonwu, et al., 2020). Our findings suggest that, among the predictors of GOIE in our study, green creativity skills exert a stronger positive influence on GOIE. This finding contradicts prior arguments for green task motivation and green expertise as the strongest drivers of GOIE (Junsheng et al., 2020; Pandita et al., 2019).

We find that green creativity skills play distinct mediating roles in the relationships between green task motivation, LSC, green expertise, environmental dynamism and GOIE. Green creativity skills are identified as a complimentary mediator—a mediating phenomenon that transfers and amplifies the positive influences of green task motivation, LSC and green expertise on GOIE. This suggests that team members who are green task motivated can further engender GOIE if they leverage their green creativity skills. Similarly, experts keen about driving green initiatives can be more efficient and effective at demonstrating relevant knowledge when green creativity skills are leveraged to drive GOIE concerns. Moreover, by leveraging green creativity skills, leaders may bolster GOIE further when driving GOIE objectives with competencies fundamental to STARA. These findings suggest that although green task motivation and green expertise have direct positive associations with GOIE, the respective green task motivation, green expertise and LSC levels of influences on GOIE could be reinforced if green creativity skills are leveraged. Our observations extend those of prior research (Awan et al., 2019; Brougham & Haar, 2018; Chang & Chang, 2017; Li et al., 2020; Liang et al., 2018; Tang & Naumann, 2016). Conversely, we find that environmental dynamism directly influences GOIE and can also negatively influence GOIE via green creativity skills. However, we further discover that green creativity skill is a competitive mediator that weakens the small negative influence of environmental dynamism on GOIE. This finding

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contradicts previous juxtapositions of environmental dynamism and GOIE in that by continuously deploying green creativity skills to explore and exploit novel pathways to combat the probable volatile negative influences of environmental dynamism on GOIE, team members can creatively predict, prepare against and foster organisational resilience against environmental dynamism effects (Huang et al., 2014; Jansen et al., 2009). This can consequently dampen environmental dynamism's small negative influence on GOIE. Therefore, LSC and green creativity skills are important actors that complement and advance the postulations of stakeholder theory (Freeman et al., 2018).

6.1 | Theoretical implications

This study is the first to attempt an empirical investigation into the long-overlooked green organisational innovation evidence (GOIE) and LSC paradigms to the best of our knowledge. We have attempted to empirically examine the emerging concept of STARA from the perspective of LSC to shed more light into how LSC influences green creativity skills and subsequently engenders GOIE. We have examined the distinct roles of green task motivation, green expertise and green creativity skills while considering the impact of a dynamic environment. Our findings extend insights into the assumptions of CTOCI (Amabile, 1997) and advance the stakeholder theory (Freeman et al., 2018) by showing that although green task motivation has the strongest positive association with green creativity skills, LSC demonstrates a higher degree of influence on green creativity skills. Although our findings complement the theorisations of CTOCI, which argues that green task motivation is positively associated with green creativity skills, we challenge such assumption by providing evidence that demonstrates LSC's greater importance in bolstering green creativity skills.

Prior research has mostly focused on how environmental dynamism influences GOIE but has overlooked a plausible association of environmental dynamism and green creativity skills. This missing insight in prior research could have helped advanced contemporary understandings and further provoked more meaningful policy implications associated with the CTOCI and stakeholder underpinnings. Consequently, we close this gap by investigating the association between environmental dynamism and green creativity skills and reporting the degree of influence environmental dynamism has on green creativity skills. We further contribute to the literature by demonstrating that environmental dynamism does have a small negative influence on GOIE. However, from our findings, it appears that environmental dynamism's negative influence on GOIE may have been greatly overrated by prior research.

Although prior research has looked into how green creativity influences the GOIE phenomenon, the components of green creativity (based on CTOCI), which could have further helped to explain how green creativity drives GOIE has long been largely overlooked. We close this gap by deepening empirical insights into how green creativity components—green expertise, task motivation and creativity skills—influence GOIE. Moreover, we provide evidence that although prior conceptualisations have championed the belief that green task motivation and green expertise are the strongest drivers of GOIE, green creativity skills do demonstrate a higher degree of influence on GOIE in our study. We consequently challenge conventional empirical extrapolations that have previously informed otherwise, thereby opening new avenues for future empirical debates.

Furthermore, we deepen insights into how green creativity skills influence the relationships between green task motivation, green expertise, LSC and environmental dynamism. We show that green creativity skills are both a complementary and competitive mediator. This is another substantial contribution to theory and practice because the dual role of green creativity skills has been overlooked by CTOCI and prior research to date. We provide support for how green creativity skills amplify the positive associations and varying degrees of influences of green task motivation, green expertise and LSC on GOIE, and how it dampens the negative influence of environmental dynamism on GOIE. These findings are of significant relevance as they challenge the CTOCI conceptualisation of the green creativity components to show further that green expertise and green task motivation are individually associated to green creativity skills. Rather than collectively act to engender OIE, the respective influences of green expertise and green task motivation on GOIE can be further enhanced by green creativity skills. We expand prior theoretical insights by demonstrating that green expertise and green task motivation exert small influences on GOIE. However, their respective small influences on GOIE can be further amplified by green creativity skills.

By demonstrating how green task motivation and green expertise influence green creativity skills, we advance contemporary insights undergirded by the CTOCI, a theory that is primarily established in psychology, human and organisational behaviour research streams. We contribute to existing research on human behaviour by deepening knowledge on the complementary and competitive roles of green creativity skills, and how it acts to amplify green task motivation, expertise and LSC on GOIE, and weaken the effect of environmental dynamism on GOIE. Given organisational behaviour research, we provide evidence that shows how the environment can impact team members' green creativity skills and GOIE. We challenge and extend existing debates that underpin the interplay between an organisation and its environment. Equally, by typifying the roles of LSC on green creativity skills and GOIE, we foster a new investigatory lens for scholars attempting to deepen insights into the interplay between leadership and STARA. The need for leaders to be competent in managing and deploying STARA to catalyse green initiatives is now paramount. We provide a fundamental praxis for future leadership, psychology, human and organisational behaviour, and the STARA research streams to engender environmental sustainability.

We also provide a novel methodological contribution by proposing new *v* effect size benchmarks in mediation analysis. To validate *v* effects size in mediation analysis, we propose a more appropriate estimate that is more approachable for *v* effect size estimations.

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6.2 | Managerial implications

To promote the achievement of the SDGs' objective of fostering environmental sustainability via increased stakeholder investments, and to remain competitive in the face of a volatile environment, leaders and policymakers ought to be aware of the special importance of green expertise, green task motivation, LSC, and environmental dynamism on green creativity skills and how they each impact GOIE. This study advocates that LSC and green creativity skills are strongly recommended prerequisites for promoting GOIE. LSC is needed to advance how team members leverage their green creativity skills to engender GOIE. Likewise, policy makers should note that, although green task motivation and green expertise are advocated to be the predominant drivers of GOIE, green creativity skills have the potency to outperform them and amplify their respective positive influences on GOIE. Consequently, even team members who are experts and are green task motivated would perform less in green initiatives than team members who exercise their green creativity skills. Leaders should thus ensure to have a workforce consisting of not only task motivated experts on green initiatives, but also team members who are trained with the capabilities of exploiting their green creativity skills for optimum performance when engaged in green initiatives. This is necessary for expediting the process of cultivating green creative ideas needed for engendering GOIE.

Resources should be allocated towards the human capital development of competencies fundamental to STARA. Given the dire uncertainty associated with radical technological advancements eclipsing the future of work, practitioners and policymakers should now, more than ever, ensure that initiatives are set up to equip team members and leaders with relevant competencies grounded in artificial intelligence, smart technology, robotics and algorithms. This may help organisations survive in the long term and help them further catalyse the objectives of the SDGs that herald the relevance of environmental sustainability. Being supported by the influence of LSC and green creativity skills, organisations would be able to more effectively and efficiently drive green initiatives, and produce green innovative evidence that can compel investors to commit towards bolstering environmental sustainability. In developing and implementing green initiatives, leaders and policymakers ought to take into consideration the small negative influences of environmental dynamism on green creativity skills and GOIE, respectively. Though the influence might be small, environmental dynamism can hamper the green creative process that could have otherwise engendered GOIE. Equally, environmental dynamism can also impede the implementation of green innovations and subsequently hinder organisations from providing green innovative evidence needed to foster stakeholder investments. Nevertheless, policymakers and practitioners can take comfort in knowing that given the support of LSC, green creativity skills could be leveraged to reduce further the already small influence environmental dynamism has on GOIE.

To further benefit from our findings, policymakers should consider exploring several resources that can aid team members in increasing their green task motivation, and develop and apply their green expertise to foster new inspirations and create novel information about their defined green initiatives. This recommendation is based on the direct positive influences of green expertise, green creativity skills and green task motivation on GOIE. Organisations should be structured in ways that inspire team members to participate in green knowledge development initiatives. Policies should be made to provoke leaders with expert STARA competencies to take up mentorship roles and help team members cultivate their own STARA competencies to drive green tasks. It is also important for team members to be given the autonomy to employ their prior green expertise to engage in difficult green tasks. This is because team members who may provide solutions to environmental sustainability problems may hesitate to do so if they feel they have no opportunity or freedom to address set problems in defined tasks. Therefore, policymakers and practitioners should ensure adequate resources are available to assist team members in executing their green tasks, especially when they need to explore diverse options in the course of utilising their green creativity skills. Moreover, team members should be given sufficient resources to support initial trials of green projects. This is important for developing, exploring and leveraging their green creativity skills for subsequent engendering of GOIE. Targeted professional green training could be initiated to enhance teams' green capabilities relevant to improving their green expertise, LSC, green task motivation, and green creativity skills.

Furthermore, with respect to our methodological contribution, we offer researchers a benchmarking proposition to adequately assess and better estimate the degree to which plausible mediators in their empirical investigations act to influence relationships between predictors and target endogenous constructs. By ascertaining exact *v* effects, researchers may arrive at more accurate conclusions that would better foster meaningful implications for policymakers and practitioners.

6.3 | Limitations and future directions

This study has been executed on a team level of analysis and grounded on the team level perspectives fundamental to advancing GOIE. Thus, individual and organisational level implications ought not to be inferred. Therefore, future researchers may attempt to advance this study's insights by investigating the study's scope from individual and organisational levels of scrutiny. This could be plausible by investigating a range of other factors related to a firm's managerial strategy, resources and quality, which may be liable to be co-determined when examining firm-level innovativeness. However, we were able to control for several established factors that can influence firm-level innovativeness. GOIE is relatively novel and might not have been sufficiently investigated in this study. We open new opportunities for future debates on the concept of GOIE. We call for researchers to initiate further qualitative and quantitative investigations to deepen insights into how organisations may best engender GOIE from an organisational perspective.

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This study applied a cross-sectional (though time lag) design that might have supressed the possibility of obtaining richer insights in executing plausible longitudinal research. Therefore, the authors call for future research to consider longitudinal investigations, especially as GOIE is relatively new and empirically underdeveloped. Future investigations may also consider investigating LSC's plausible influences and environmental dynamism on green task motivation and green expertise as that was not examined in our study. Based on prior research, LSC's possible moderating influences and environmental dynamism on the relationships between green task motivation, green expertise, and green creativity skills on GOIE present another exciting area for further investigations. In order to stay within the scope of our aims, we did not pursue this.

Furthermore, caution should be taken when generalising this study's results. Data employed for this study's analysis emanated from only 55 manufacturing organisations, which are nevertheless nationally recognised and reflect this study's core objectives. Moreover, future research should endeavour to extend investigation to include more organisations by reaching into other industries within and across Nigeria (or elsewhere). Finally, although we framed this study to better understand GOIE to attract and retain investor interest, we did not actually collect data or analyse investor activity. We encourage future research studies to expand on these matters.

CONFLICT OF INTEREST

No potential conflict of interest was reported by the authors.

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ENDNOTE

[1] Despite proposing a new calculator for v effect sizes, our indirect effects did not reach this more reachable target. Although this is unfortunate, it is simply a consequence of our model and data, and we hope it is not seen as undermining our recommendation for the new *v* thresholds.

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APPENDIX A.

Green expertise measurement items:

1. This team member can address almost any problem in his/her green related tasks.
2. This team member is considered an expert in his/her area of green related tasks
3. In tackling green job difficulties, this team member is preferred to other colleagues.
4. This team member offers constructive feedbacks about problems from his/her green related tasks.

Green creativity skills measurement items:

1. This team member is highly skilled in thinking up new ways of driving green initiatives.
2. This team member is highly skilled at generating more than one solution to a problem associated with green initiatives.
3. This team member is highly skilled in finding out the root cause of a problem in a green related task.
4. This team member is highly skilled in being able to pick the best option from a number of solutions to a green related problem.

Green task motivation measurement items:

1. I am strongly motivated by the recognition I can get from my company for engaging in a green initiative.
2. I like to set myself challenging targets in my green-related task.

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3. I enjoy doing new things in my green related task.
4. When things seem difficult, I make a special effort to keep trying to drive green initiatives related to my job.
5. I am not very concerned how things turn out in my green-related task (reverse coded).

Leader STARA competency measurement items:

1. My leader has the knowledge and ability to apply smart (analysing, reporting and self-monitoring systems) technology during operations.
2. Matters related to machines that share similar qualities (learn, reason, discover and calculate) with the human mind are adequately addressed by my leader.
3. My leader is not good at designing or applying algorithms to complete defined tasks (reverse coded).
4. My leader knows how to design and apply robots or mechanical devices during operations.

Green organisational innovative evidence measurement items:

1. My organisation currently owns one or more patented inventions that can positively contribute to environmental sustainability.
2. My organisation engages in the exports of high green technology products or services.
3. My organisation's green products or services are ISO 14001 certified.

Environmental dynamism:

1. In our market, nothing has changed in 1 year (reverse coded).
2. In our local market, environmental changes are intense.
3. There are regular requests for new services and product by our clients.
4. There are continuous changes in our local market.
5. There is a rapid and frequent change in the volume of services and products in our market.

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